

MONROE CUSTER, MI, USA

WMO: 725418

Lat: 41.940N

Lon: 83.435W

Elev: 612

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54823

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.8	7.0	-7.5	3.8	1.2	-0.2	5.5	8.8	26.0	27.5	23.9	26.6	5.8	240	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	90.4	74.2	88.0	73.4	84.3	71.8	77.9	85.2	76.0	83.2	74.5	81.2	10.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	136.0	81.3	73.4	127.2	78.9	72.5	123.0	78.2	41.7	85.1	39.8	83.2	38.4	82.3	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.1	19.5	17.3	-5.5	94.1	6.3	2.5	-10.0	96.0	-13.7	97.4	-17.2	98.9	-21.8	100.7	
			-6.1	80.7	6.0	2.1	-10.5	82.2	-14.0	83.5	-17.4	84.6	-21.8	86.2	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	26.1	27.9	36.9	48.8	60.3	69.7	73.0	70.8	64.8	53.1	41.9	31.4
	DBStd	18.77	11.47	10.68	10.26	8.83	8.45	6.61	5.60	5.18	7.18	8.65	9.22	10.05
	HDD50	2844	742	621	425	124	11	0	0	0	1	65	275	580
	HDD65	6052	1206	1039	871	490	198	28	4	10	91	381	693	1041
	CDD50	3031	1	2	19	89	330	590	712	646	444	162	32	4
	CDD65	764	0	0	0	5	51	168	251	191	85	13	0	0
	CDH74	6692	0	0	8	72	476	1485	2324	1540	691	95	1	0
	CDH80	2035	0	0	0	8	130	504	783	403	190	17	0	0
Wind	WSAvg	7.1	8.8	8.4	8.1	8.6	7.2	6.4	5.2	4.6	5.1	6.8	7.6	8.3
Precipitation	PrecAvg	33.3	2.0	1.9	2.4	3.3	3.4	3.5	3.3	3.1	2.4	2.6	2.3	
	PrecMax	46.9	3.6	5.1	5.5	5.0	5.8	8.0	6.4	7.9	6.2	6.1	6.6	6.0
	PrecMin	25.3	0.6	0.4	0.6	1.1	0.8	1.7	1.3	0.6	0.6	0.5	0.8	0.6
	PrecStd	4.9	0.9	1.1	1.1	1.1	1.6	1.4	1.1	1.5	1.4	1.3	1.2	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.7	62.1	72.8	81.3	88.2	92.8	93.3	90.7	90.3	82.1	70.2	60.9
		MCWB	55.6	54.4	62.3	64.6	70.6	72.9	76.1	75.6	74.0	68.1	59.2	57.3
	2%	DB	52.3	54.2	63.8	75.0	83.7	90.0	90.4	87.9	85.1	75.9	64.1	54.5
		MCWB	50.1	50.6	56.3	60.9	68.8	72.4	75.1	74.8	71.5	65.1	56.7	51.5
	5%	DB	45.9	46.2	57.5	70.2	80.4	85.9	87.9	84.0	81.3	72.2	60.8	48.4
		MCWB	43.0	42.5	51.6	58.6	67.1	71.0	74.2	73.3	69.4	62.8	54.3	45.7
	10%	DB	40.8	41.1	53.3	64.0	74.8	81.9	84.1	81.7	78.2	68.0	55.5	44.9
		MCWB	38.3	37.6	48.2	55.0	64.9	68.9	72.6	71.0	68.0	61.1	50.7	42.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.2	56.5	63.5	67.2	74.5	77.3	80.9	79.4	77.8	72.5	62.5	57.4
		MCDB	58.3	59.8	70.7	76.8	82.4	87.2	86.7	86.3	85.6	78.4	66.9	60.5
	2%	WB	50.6	50.3	57.9	63.6	71.6	75.3	78.5	77.3	74.8	68.3	58.7	52.4
		MCDB	52.0	54.1	63.1	70.8	79.7	84.0	85.7	83.4	81.2	73.2	63.2	54.5
	5%	WB	43.5	42.9	53.3	59.9	69.1	73.7	76.4	75.4	72.4	64.8	55.1	46.3
		MCDB	45.7	46.0	57.2	68.3	76.6	81.5	83.9	81.3	78.1	69.3	59.1	48.0
	10%	WB	38.3	37.9	47.7	56.3	66.7	72.0	74.6	73.6	70.2	61.8	51.9	42.1
		MCDB	40.5	40.5	52.3	64.0	73.3	79.3	81.4	79.2	75.2	66.8	55.7	44.7
Mean Daily Temperature Range	5% DB	MDBR	13.2	14.0	15.9	18.7	19.0	19.0	19.7	19.2	20.4	18.7	15.2	11.8
		MCDBR	16.8	19.8	24.2	26.9	23.7	22.9	21.9	21.3	24.0	23.3	21.4	16.5
		MCWBR	15.2	16.8	17.5	15.8	11.9	10.0	9.7	10.7	12.3	13.3	15.9	14.6
	5% WB	MCDBR	16.2	19.3	23.2	23.6	20.5	19.3	18.9	18.0	19.1	19.2	19.4	15.5
		MCWBR	15.3	17.3	18.1	16.2	12.2	10.6	9.9	9.9	11.2	12.9	16.1	14.7
Clear-Sky Solar Irradiance	taub	0.301	0.337	0.358	0.404	0.435	0.452	0.443	0.435	0.389	0.361	0.322	0.314	
	taud	2.407	2.287	2.324	2.225	2.186	2.186	2.231	2.241	2.346	2.406	2.478	2.401	
	Ebn at Noon	268	272	281	274	268	263	264	262	267	262	259	251	
	Edh at Noon	25	33	36	43	46	46	43	42	35	29	23	23	
All-Sky Solar Radiation	RadAvg	486	769	1111	1465	1714	1919	1928	1681	1363	894	565	392	
	RadStd	48	99	105	125	148	126	102	63	115	75	75	36	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+1.40	N/A	N/A	N/A	N/A	
(47)	Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+155	N/A	

Nomenclature: See separate page